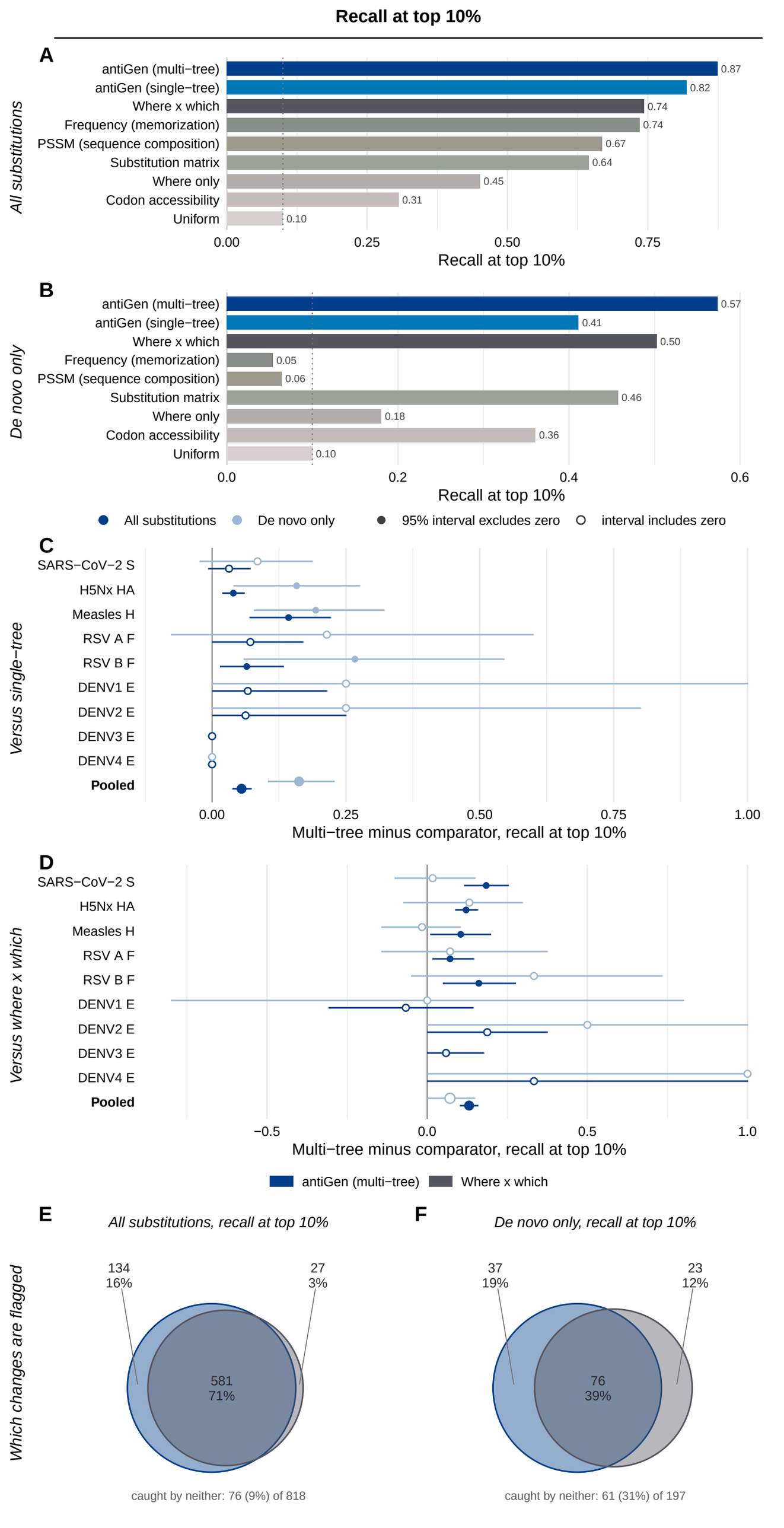
Recall at top 10%
A
All substitutions
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
PSSM (sequence composition)
Substitution matrix
Where only
Codon accessibility
Uniform
0.87
0.82
0.74
0.74
0.67
0.64
0.45
0.31
0.10
0.00
0.25
0.50
0.75
Recall at top 10%
B
De novo only
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
PSSM (sequence composition)
Substitution matrix
Where only
Codon accessibility
Uniform
0.57
0.41
0.50
0.05
0.06
0.46
0.18
0.36
0.10
0.0
0.2
0.4
0.6
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
0.00
0.25
0.50
0.75
1.00
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 10%
Which changes are flagged
134
16%
27
3%
581
71%
caught by neither: 76 (9%) of 818
F
De novo only, recall at top 10%
37
19%
23
12%
76
39%
caught by neither: 61 (31%) of 197